MILITARY SPECIFICATION

PUMP, VACUUM, POSITIVE DISPLACEMENT, ELECTRIC MOTOR DRIVEN

1. SCOPE

1.1 This general specification covers pumps, vacuum, positive displacement type, electric motor driven.

1.2 The performance parameters shall be as specified in section 6.2 of this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-636	Motor, Alternating-Current (Fractional Horsepower)
CC-M-641	Motor, Alternating-Current (Integral Horsepower)
TT-T-656	Tricresyl Phosphate
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-T-60	Tape, Pressure Sensitive, Adhesive, Waterproof for Packaging

Military

MIL-P-116	Preservation, Methods Of
MIL-C-3432	Cable, Power Supply
MIL-C-6529	Corrosion Preventive Compound
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-E-7729	Protective Coatings

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

MIL-STD-130	Identification Marking
MIL-STD-143	Specifications and Standards
MIL-STD-810	Environmental Test Methods
MIL-STD-1186	Cushioning, Anchoring, Blocking, Bracing and Waterproofing Test Methods

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or the request for proposal shall apply.

American Society of Mechanical Engineers Power Test Code

19.5.4 Supplement on Instruments and Aparatus - Part 5, Chapter 4. Flow measurement by means of standardized nozzles and orifice plates.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York 17, N.Y.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The vacuum pump shall consist of the following major components:

Description	Item
Vacuum Pump	1
Electric Motor	2
Accessories	3

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a suitable fungicidal agent.

3.4.2 Metals. Metals shall be of the corrosion resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

3.5 Design and construction.

3.5.1 The vacuum pump shall have a single inlet and shall be a self contained unit with pump, drive, and accessories mounted to a common base.

3.5.2 The design shall provide for the least number of selectively fitted component parts.

3.5.3 The vacuum pump design shall incorporate a minimum number of component parts consistent with the performance specified herein.

3.5.4 The design life should be 4000 hours minimum.

3.6 Performance. The vacuum pump shall be capable of meeting the requirements specified herein. See MIL-STD-810B for definition environmental conditions.

3.6.1 Capacity. The free air capacity of the vacuum pump shall be such that when it is operating at rated RPM under standard ambient conditions the quantity of air pumped shall be as specified in 6.2.b.

3.6.2 Ultimate vacuum. When operating at rated speed under standard ambient conditions with the inlet blanked off the vacuum pump shall attain the absolute pressure specified in para 6.2.d.

3.6.3 Pumping speed. The pumping speed shall be such that when operating at rated speed at standard conditions, the evacuation time shall not exceed that specified in para 6.2.c.

3.6.4 Operation. The vacuum pump shall be capable of operating continuously under full load or no load conditions.

3.7 Details of components.

3.7.1 Separator. The vacuum pump shall be provided with a device that will recover the lubricating fluid from the discharge gases, separate condensible vapors from the lubricating fluid, and provide for discharging the condensible vapors from the pump (if specified in para 6.2.h).

3.7.2 Gas ballast. The vacuum pump shall be equipped with a gas ballast system to prevent the condensation of vapors in the oil (if specified in para 6.2.h).

3.7.3 Lubricant. The vacuum pump shall operate in accordance with the requirements specified herein when lubricated with commercial grade vacuum pump oil as specified in 6.2.m or tricresyl phosphate conforming to TT-T-656.

3.7.3.1 Lubricant level. The vacuum pump shall be provided with a lubricant level indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

3.7.4 Inlet connection. The vacuum pump inlet and outlet connection shall be as specified (para 6.2.f).

3.7.5 Check. The vacuum pump shall be provided with a device that will prevent flow of the vacuum pump lubricant into the container being evacuated if the pump stops (if specified para 6.2.k).

3.7.6 Cooling. The vacuum pump unit shall have an air or water cooling system of adequate capacity to maintain the pump at proper operating temperatures.

3.7.7 Electric motor. The electric motor shall conform to CC-M-636 or CC-M-641 as applicable, with enclosure as specified in para 6.2.g.

3.7.7.1 Motor size. The electric motor shall be sized to drive the load such that the NEMA motor ratings are not exceeded.

3.7.7.2 Electric power. The voltage and frequency shall be as specified in para 6.2.g.

3.7.7.3 Control voltage. The control voltage shall not exceed 120V.

3.7.7.4 Power supply cable. The power supply cable shall conform to MIL-C-3432 and as specified in para 6.2.g.

3.7.8 Controls. The vacuum pump unit shall be equipped with all controls necessary for ready starting and stopping of the unit.

3.7.8.1 Automatic controls. Automatic safety devices shall be provided as needed to insure safety of personnel and equipment.

3.8 Commercial item. The vacuum pump unit shall be a standard item modified only to meet the requirements of this specification. The unit shall be completely assembled and mounted to a rigid base ready to operate.

3.9 Protective caps. Protective caps shall be provided to prevent moisture and foreign material from entering the vacuum pump during transport or storage.

3.10 Tool compartment. When specified in para 6.2.i, a compartment suitable for storage of special tools shall be provided.

3.11 Dimensions. The overall dimensions of the pump unit shall not exceed the envelope limits as specified in 6.2.j.

3.12 Guards. Moving component parts that are a possible hazard to personnel shall be enclosed or otherwise guarded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

3.13 Protective coatings. Exterior surfaces of this vacuum pump assembly shall be finished with two coats of gray gloss enamel, color 512 conforming to MIL-E-7729.

3.14 Identification of product. Assemblies shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the vacuum pump unit shall be classified as follows:

a. Preproduction tests----------see 4.4.
b. Acceptance tests----------see 4.5.

4.3 Test conditions.

4.3.1 Test conditions shall conform to the requirements of MIL-STD-810B, para 3.1.

4.3.1.1 Pressure. Absolute pressures shall be measured with equipment that indicates the total pressure.

4.3.1.2 Temperature. Temperatures shall be expressed in fahrenheit units.

4.3.2 Airflow. Airflow for free-air displacement tests shall be measured with standardized nozzles or orifices as specified in ASME Code 19.5:4. The rate of air flow shall be recorded in terms of the actual volume occupied under the existing ambient conditions.

4.4 Preproduction tests.

4.4.1 Test sample. One new vacuum pump shall be subjected to the tests specified in para 4.6. Failure to meet these tests shall be cause for rejection.

4.5 Acceptance tests. Acceptance tests shall consist of individual tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

4.5.1 Individual tests. Each vacuum pump unit shall be run in and subjected to the tests specified in para 4.6.1, 4.6.2, 4.6.3, (4.6.3.1 not required), 4.6.6.1, 4.6.6.2, and 4.6.6.3. Failure to meet these tests shall be cause for rejection.

4.6 Test methods.

4.6.1 Examination of product. The vacuum pump shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Functional check. Accessory mechanical assemblies, if any, shall be checked to insure proper operation.

4.6.3 Operational check. The vacuum pump shall exhibit no evidence of malfunction when subjected to the following tests:

4.6.3.1 Proof running. The vacuum pump shall undergo an operational proof run time of not less than 8 hours. The proof time shall be divided as follows and in the order listed.

a. Four hours of continuous operation at rated speed with the vacuum pump inlet and discharge connections open to atmospheric pressure.

b. Four hours of continuous operation at rated speed with the vacuum pump inlet completely blanked off except for a connection to the absolute pressure measuring device. The absolute pressure attained at the inlet shall be recorded.

4.6.4 Performance checks. The vacuum pump shall be subjected to the following performance checks.

a. Free air pumping rate.

b. Pumping rate at low pressures.

4.6.5 Initial teardown inspection. The vacuum pump used for testing shall be disassembled and inspected prior to further testing. All wearing parts and all parts subjected to high stresses in operation shall be accurately measured, ferrous parts shall be subjected to magnetic particle inspection in accordance with MIL-I-6868, and nonmagnetic parts shall be inspected with fluorescent penetrant (black light) in accordance with MIL-I-6868 for comparison with like inspections to be performed after completion of the tests. The vacuum pump shall be reassembled and subjected to the remainder of the tests.

4.6.6 Performance demonstration. The vacuum pump shall be tested in accordance with the following procedures to determine its performance capabilities prior to endurance testing. Any malfunctions or unusual operations difficulties occuring during the tests shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

4.6.6.1 Free air pumping rate. The vacuum pump shall be operated at rated speed until its free air pumping rate (see para 6.2.b) is determined.

4.6.6.2 Ultimate vacuum. The vacuum pump inlet shall be completely blanked off except for a connection to the absolute pressure measuring device and operated at rated speed with the discharge open to atmospheric pressure until the inlet pressure has remained stable for not less than ½ hour. The ultimate vacuum attained shall be as specified in 6.2.d.

4.6.6.3 Pumping speed. The pumping rate at reduce pressure shall be demonstrated and shall meet the requirement of para 3.6.3.

4.6.6.4 Test with tricresyl phosphate. If tricresyl phosphate is specified in 6.2m the tests specified in 4.6.6.1, 4.6.6.2 and 4.6.6.3 shall be repeated using tricresyl phosphate.

4.6.7 Endurance. The vacuum pump shall be subjected to an endurance test of not less than 480 hours. The lubricant and sealing fluid used throughout this test shall be as specified in 6.2.m. The endurance test time shall be divided as follows:

a. One hundred twenty hours of continuous operation with the vacuum pump inlet pressure between 500 mm hg. absolute and atmospheric pressure.

b. Two hundred forty hours of continuous operation with the vacuum pump inlet pressure at fifty percent ultimate pressure.

c. One hundred twenty hours of continuous operation with the vacuum pump inlet completely blanked off except for a connection to an absolute pressure measuring device and the discharge open to atmospheric pressure.

4.6.8 Environmental testing. If environmental testing is required (6.21) the vacuum pump unit shall be subjected to the following tests conducted in conformance with MIL-STD-810.

4.6.8.1 Low temperature test. The vacuum pump unit shall be subjected to the low temperature test specified in procedure I of method 502. The vacuum pump unit shall be subjected to an ambient temperature of -65°F for not less than 24 hours. The vacuum pump shall then be started in accordance with the manufacturers instructions and operated with the inlet blanked off until the inlet pressure reaches the ultimate vacuum. Upon completion of the low temperature tests the vacuum pump shall be examined for damage.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:54Z
Check the source to verify that this is the current version before use.

4.6.8.2 High temperature test. The vacuum pump unit shall be subjected to the high temperature test specified in procedure I of method 501. The vacuum pump unit shall be operated at the ultimate vacuum in an ambient temperature of $125^{\circ} \pm 5$ F for not less than 48 hours. Upon completion of the high temperature test the vacuum pump shall be examined for damage.

4.6.8.3 Humidity test. The vacuum pump shall be exposed to, but not operated in, humidity in accordance with procedure I.

4.6.8.4 Fungus test. The vacuum pump shall be exposed to, but not operated in, fungus in accordance with procedure I.

4.6.8.5 Salt-fog test. The vacuum pump shall be exposed to, but not operated in, salt-fog in accordance with the procedure specified.

4.6.9 Shock tests.

4.6.9.1 Impact. The vacuum pump shall be propelled at a speed of not less than 5 mph into a solid, fixed wall. This test shall be repeated until all four sides of the vacuum pump have been subjected to the impact. Upon completion of the test, the vacuum pump shall be operated and any leaks, component malfunctions, et cetera, shall be recorded in detail on test data sheets.

4.6.9.2 Drop test. The vacuum pump shall be raised to a height of not less than 6 in. and dropped onto a solid, concrete floor. The vacuum pump shall then be operated and any operating discrepancies shall be reported on the test data sheets.

4.6.10 Servicing and maintenance tests. All normal preventive maintenance and servicing operations specified in the operating and maintenance instruction handbook shall be performed to determine their adequacy, ease of accomplishment, and accessibility of parts and assemblies for performance of same. Insofar as practicable, these tests shall be conducted as part of the normal preventive maintenance, servicings, and inspections performed in accomplishing the testing specified herein. Interferences or obstructions to servicing or preventive maintenance shall be recorded in detail on the test data sheets.

4.6.11 Final teardown inspection. The vacuum pump shall be disassembled as specified in 4.5.6 and a critical inspection made of all components to determine their operability and any damage or undue wear encountered during the testing.

4.6.11.1 The measurements taken as specified in 4.5.6 shall be repeated for comparison with the data recorded prior to testing. Wear or distortion that exceeds the limits established by the manufacturer for new parts shall be cause for considering the affected parts as having failed to satisfactorily pass the tests.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:54Z
Check the source to verify that this is the current version before use.

4.6.11.2 Ferrous parts shall be resubjected to magnetic particle inspection (magnaflux) in accordance with MIL-I-6868 and shall exhibit no indications of damage attributable to the tests, or they shall be considered as having failed the tests. Grind checks and subsurface indications of laps, seams, or other forging defects noted before the test and that have not increased in size, shall not be cause for rejection.

4.6.12 Nonmagnetic casting, forgings, and other operating parts shall be subjected to inspection with fluorescent penetrant (black light) in accordance with MIL-I-6866, and shall exhibit no cracks or other defects, or shall be considered to have failed the tests.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Levels A or C as specified. (See 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Prior to any preservation and packaging, all oil shall be drained from the vacuum pump. External surfaces shall be thoroughly cleaned and dried in accordance with the applicable process of MIL-P-116.

5.1.1.2 Each vacuum pump shall be packaged Method II in accordance with MIL-P-116.

5.1.2 Level C. Vacuum pumps shall be preserved and packaged in accordance with the manufacturers commercial practice.

5.2 Packing. Packing shall be Levels A, B or C as specified. (See 6.2)

5.2.1 Level A. Vacuum pumps when preserved and packaged in accordance with 5.1 thru 5.1.1 shall be packed in an exterior container, overseas type, Style A conforming to PPP-B-601. Containers shall be of uniform size and shape, of minimum cube and tare consistant with the protection required. Containers shall be closed and strapped in accordance with the container specification or appendix thereto.

5.2.2 Level B. Same as Level A, except container shall be of domestic type.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:54Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be packed in an exterior container in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing and bolting shall be in accordance with MIL-STD-1186.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Vacuum pump is intended for use in applications where there is a requirement to evacuate air from a finite space.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Free air pumping capacity. _____ CFM minimum (3.6.1).
c. Pumping speed; time to reduce pressure in a _____ volume from atmospheric pressure to _____ absolute pressure not to exceed _____. (3.6.3)
d. Ultimate vacuum (3.6.2).
e. Inlet connection length and diameter (3.7.4). (If ultimate vacuum below 10 militorr.)
f. Inlet and outlet size and configuration (3.7.4).
g. Electric power requirements, enclosure, and power supply cable length (3.7.7).
h. If separator and or gas ballasting required (3.7.1).
i. Tool compartment required (3.10).
j. O.A. envelope. Dimensions if required (3.11).
k. Check valve (3.7.5).
l. Environmental testing required (4.6.8).
m. Pump lubricant (3.7.3).
n. Applicable levels of preservation, packaging and packing (see 5.1).

Custodian:
Air Force - 84

Preparing Activity:
Air Force - 84

Project Number:
4310-F020

563

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. 67 AFLC RAFB,GA JUN 67 • 600

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

FOLD

WRAMA (WRNEME)
ROBINS AFB GA 31093

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

WRAMA (WRNEME)
Robins AFB GA 31093

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.